

# TRIVANTAS



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**LEVEL SWITCHES**



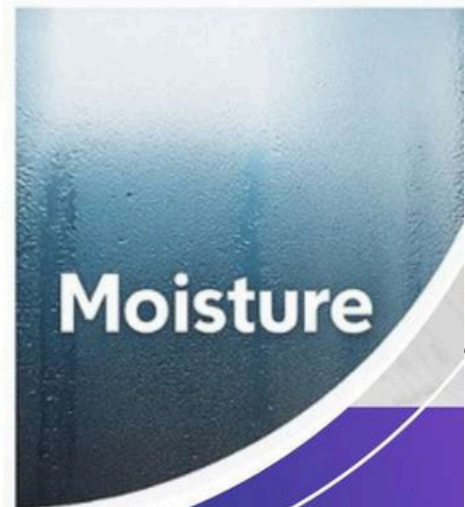
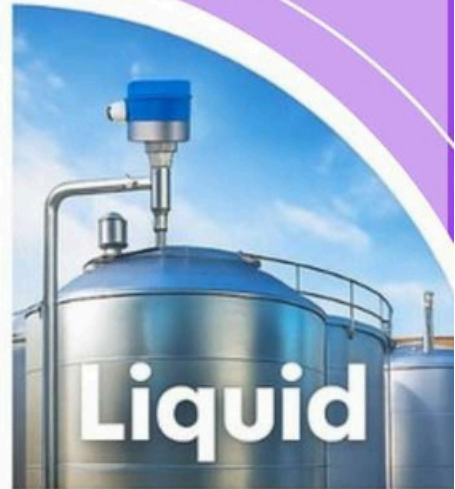
**LEVEL TRANSMITTERS**



**LEVEL TRANSMITTERS**



**MOISTURE SENSORS**



TRIVANTAS

SMART LEVEL SENSORS | ADVANCED FILTRATION SOLUTIONS

MATERIAL HANDLING EQUIPMENT | SPM

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# VIBRATING FORK LEVEL SWITCH :

## Technical Specification :

|                       |                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------|
| Housing               | : Steel/ Aluminium, weatherproof ,powder coated ,<br>integral with the probe                     |
| Cable Entry           | : 2 no.s (in Cast Al. Housing)                                                                   |
| Ambinent temperate    | : 0°C to + 60°C                                                                                  |
| Power consumption     | : 0.3 VA                                                                                         |
| Mains Voltage         | : 18 to30 V DC or 65 to 265 V AC (Cast Al. Hou.) std.                                            |
| Output                | : PNP or relay output in Cast Al. Housing.                                                       |
| Switching delay       | : Continuously adjustable From 1 to 20 sec.<br>probe free or probe covered (in cast Al Version). |
| Safety operation      | : Field selected switch over min. or max (In Cast Al Version).<br>(FSL / FSH) switching points.  |
| Switch status display | : Green LED shows Normal, Red LED shows alarm.                                                   |



## FORK (SENSING PROBE) :

|                   |                                                                |
|-------------------|----------------------------------------------------------------|
| Mounting          | : Screwed – 1 1/2” BSP (standard) or flanged (optional)        |
| Sense             | : Fork, Stainless steel                                        |
| Extension         | : Pipe (optional) in extended type                             |
| Length            | : 200 mm min. up to 3000 mm and as per application             |
| Operating Temp in | : 150°C for the integral switching vessel unit. 200°C optional |

## Models :

|             |                                                                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TS 2010     | : 120hz frequency, weatherproof IP 67 cast Al housing with<br>universal mains.18-30 VDC and 65 to 265 VAC. 2 x SPDT relay output                                                |
| TS 2020     | : 120hz frequency, weatherproof IP 67 cast Al compact housing<br>PNP DC output.                                                                                                 |
| TS 2030     | : 120hz frequency, weatherproof IP 68 C steel Pipe housing.<br>PNP DC output.                                                                                                   |
| TS 2020 Ex. | : 120hz frequency, weatherproof IP 65. Flameproof as<br>per BIS standard vide CIMFRTc/sR/H642 DT 16tA2t20<br>version) and PESO certificate(In PNP version) and PESO certificate |

## VIBRATING FORK LEVEL SWITCH

### Technical Specification :

|                      |                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------|
| Housing              | : Steel/Aluminium, weatherproof , powder coated, integral with the Probe                    |
| Cable Entry          | : 1 no. s (Cast Al. Housing)                                                                |
| Ambient temperature. | : 0°C to +60°C                                                                              |
| Power consumption.   | : 0.3 VA                                                                                    |
| Mains Voltage.       | : 18 to 30 VDC or 65 to 265 VAC std.                                                        |
| Output               | : ENP or DPDT Tin cast Al. Housing                                                          |
| Switching delay.     | : Continuous adjustable from 1 to 20 sec. probe free or probe covered (in cast Al version). |

### Output : ENP or DPDT

Safety operation : Field selected switch over for (In Cast Al Version) min or max  
Switch status display : Green LED shows Normal, Red LED shows alarm.

### FORK (SENSING PROBE) :

|                     |                                                                             |
|---------------------|-----------------------------------------------------------------------------|
| Mounting            | : Screwed – 1" BSP (standard) or flanged (optional)                         |
| Sense               | : Fork, Stainless steel                                                     |
| Extension           | : Pipe (optional) in extended type                                          |
| Length              | : 70 mm (standard) up to 3000 mm and as per application                     |
| Operating Temp in : | 150°C for the integral switching vessel unit.<br>(standard)/ 200 C optional |

**APPLICATION :** Most liquids with max 10000 mm<sup>2</sup>/s viscosity.

### MODELS :

|             |                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| TS 7000     | : 2Khz frequency, weatherproof IP 67 cast Al housing with universal mains. 18-30 VDC and 65 to 265 VAC. 1 DPDT relay output |
| TS 7220     | : 2Khz frequency, weatherproof IP 67 cast Al compact housing. PNP DC output.                                                |
| TS 7230     | : 2Khz frequency, weatherproof IP 68 C steel Pipe housing. PNP DC output.                                                   |
| TS 7200 Ex. | : 2Khz frequency, weatherproof IP 65. Flameproof as per BIS standard<br>(In PNP version) and PESO certificate               |



# VIBRATING FORK LEVEL SWITCH

## TECHNICAL SPECIFICATION :

|                       |                                                                        |
|-----------------------|------------------------------------------------------------------------|
| Housing               | : Steel/Aluminium, weatherproof Powder coated, integral with the probe |
| Cable Entry           | : 1 no.s (standard)                                                    |
| Ambient temperature   | : 0°C to + 60°C                                                        |
| Power consumption     | : 0.3 VA                                                               |
| Mains Voltage         | : 18 to 30 VDC (In Cast Al. Housing)                                   |
| Output                | : PNP & DPDT in Cast Al. Housing                                       |
| Switch status display | : Green LED shows Normal, Red LED shows alarm.                         |

## FORK (SENSING PROBE) :

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| Mounting          | : Screwed 1/2" / 3/4" BSP (standard) or flange(Optional) |
| Sense             | : Fork, Stainless steel                                  |
| Extension         | : Pipe (optional) in extended type                       |
| Length            | : 70 mm (standard) up to 3000 mm and as per application  |
| Operating Temp in | : 150°C for the integral switching vessel unit.          |
| Vessel            | : 200°C optional                                         |
| Pressure          | : Upto 40 Bar (Inside vessel)                            |

**APPLICATIONS :** Most liquides with max 10000 mm<sup>2</sup>/s viscosity.

## Models :

|            |                                                                                         |
|------------|-----------------------------------------------------------------------------------------|
| TS 7110    | : 1.8khz frequency, weatherproof IP 67 cast Al housing                                  |
| TS 7130    | : 1.8khz frequency, weatherproof IP 68 C steel Pipe housing<br>PNP DC output.           |
| TS 7100 EX | : 1.8 khz frequency, weatherproof IP 65. Flameproof as<br>per BIS and PESO certificate. |



## VIBRATING FORK LEVEL SWITCH :

### Technical Specification :

#### SWITCHING UNIT :

|                     |                                                                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Housing             | : Steel/Aluminium, weatherproof , powder coated, integral with the probe<br>Powder coated. Integral with the probe Steel / Aluminum, weather proof |
| Cable entry         | : 1 nos. (Cast Al. Housing)                                                                                                                        |
| Ambient temperature | : 0 ° C to +60° C                                                                                                                                  |
| Power consumption   | : 0.3VA                                                                                                                                            |
| Mains Voltage       | : 18 to 36 V DC                                                                                                                                    |
| Output              | : PNP or Relay output in cast Al. Version .                                                                                                        |
| Switching delay     | : Continuously adjustable From 1 to 255 sec.                                                                                                       |
| Safety operation    | : Probe free or probe covered. Field selected switch over min. or max<br>Status (In Cast Al Version). (FSL / FSH) switching points.                |
| Display             | : Green LED shows Normal, Red LED shows alarm.                                                                                                     |



#### FORK (SENSING PROBE) :

Mounting : Screwed – 1 ” BSP (standard) Or Flanged (optional)  
Sense : Fork, Stainless steel , Pipe ( optional) in extended type  
Extension : Pipe ( optional) in extended type  
Length : 125 mm (standard) up to 3000mm and as per application.  
Operating Temp : 150 deg for the integral switching Vassal unit (standard) 250° C Optional.  
Pressure : upto 40 Bar (Inside vessel)

**APLLICATIONS :** Most liquids with max 10000 mm<sup>2</sup>/s viscosity. Corrosive, thick, turbulent, flowing liquids. Powders and light granules with min. density 0.05 kg/dm<sup>3</sup>.

#### Model :

- TS 7010 : 330hz frequency, Weather proof IP 67 Cast Al housing. With universal mains.  
18- 30 V Dc and 65 to 265 V AC. 2 x SPDT relay output.
- TS 7020 : 330hz frequency, Weather proof IP 67 Cast Al compact housin. PNP DC output
- TS 7020 Ex : 330Hz frequency, Weather proof IP 65, Flameproof as per BIS and PESO certificate
- TS 7020 : 330hz frequency, Weather proof IP 68 Steel Pipe housing. PNP DC output
- TS-7040 : 330 Hz frequency, Weatherproof IP69K, Stainless Steel housing, PNP/NPN switchable outpu

## RF/NRF Admittance Level Switch : SWITCHING UNIT :

### Technical Specification : (microcontroller based)

|                       |                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------|
| Housing               | : Steel/Aluminium, weatherproof ,powder coated ,<br>integral with the probe                    |
| CableEntry            | : 1no.s(CastAl. Housing)                                                                       |
| Ambinent temperate    | : 0°C to + 60°C                                                                                |
| Power consumption     | : 0.3 VA                                                                                       |
| MainsVoltage          | : 18to30VDCor 65 to 265 V AC (Cast Al. Housing) std.                                           |
| Output                | : PNP or relay output in Cast Al. Housing.                                                     |
| Switching delay       | : Continuously adjustable From 1 to 20 sec.<br>probe freeorprobe covered (in cast Al Version). |
| Safety operation      | : Field selected switch over min. or max (In CastAlVersion).                                   |
| Switch status display | : GreenLEDshows Normal, Red LED shows alarm.                                                   |
| Operating Frequency   | : 6 KHz                                                                                        |
| sensitivity Setting   | : Digitally setable                                                                            |

### COAT GUARD SENSING PROBE :

|                       |                                                                                |
|-----------------------|--------------------------------------------------------------------------------|
| Mounting              | : Screwed – 1 ½” BSP (standard) Or Flanged (optional) or as per specification. |
| Sense rod             | : Stainless steel                                                              |
| Shield                | : Stainless steel                                                              |
| Insulation            | : PTFE (standard)Other on request and as per application.                      |
| Operating temperature | : 200 ° C max.                                                                 |
| Pressure              | : upto 60 Bar (Inside vessel)                                                  |

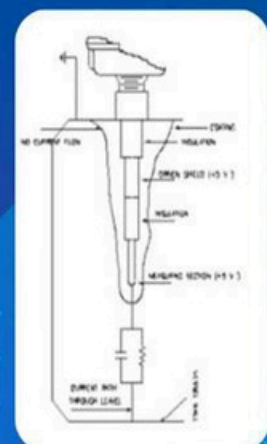
### THEORY :

The RF level switch operates on the basis of RF absorption measurement. The electronic unit generates a sinusoidal wave, applied to the electrode creating a field around it. RF environment absorption changes (electrical loss) around the electrode are reflected on changes of generator supply current. Such changes, caused by the increase in level is amplified and used to energise the relay. The main drawback of the conventional method is that after the level has once increased and then decreased, there may be a coating left on the probe which is sensed by the instrument as though the level is still on the probe. In RFADMIT a COAT GUARD and Trance Conductance amplifier are incorporated in the circuit having its output exactly at the same voltage and phase at all times as its input. The output is connected through the shield of the low capacitance co-axial cable to the concentric tube on the sense probe, called shield element. Since both the elements, sense and shield are exactly at the same potential and phase at all times, there is no current flow through the cable. Thus there is no change in calibration due to coating on the probe and the temperature effect of the cable.

Models Available in NRF:

RFADMITT 3600 : With PNP output , DC mains 18 to 30V DC

RFADMITT 3610 : With DPDT Relay Output, Manins 18-36V DC & 65 to 265VAC



## CAPACITANCE LEVEL POSITION DETECTOR : PiCap

**PiCap** is capable of detecting metallic and non-metallic objects at long distances, reliably and efficiently. **PiCap** sensitivity is factory set to detect most of the available substances in common Industrial applications. Since the detection range depends on the dielectric constant of the media and the nearness of objects surrounding the installation in special cases, the sensitivity may need adjustment.



### Features :

- Available in Stainless steel housing
- Flush or non-flush types
- Long sensing range
- Multi-turn Pot provided for sensitivity adjustment in typical applications
- Current limited open collector PNP transistor output as standard
- Reverse polarity and transient voltage protected
- Selectable Fail-safe High or Low
- Built-in LED indicators for Status, Alarm and Normal operation

### Where used ?

PiCap can detect almost all types of conductive and non-conductive materials, such as water, glass, plastics, and metals without physical contact.

## Typical Applications

- Grain and plastic pellet detection
- Object detection and counting
- Liquid level detection in non-metallic containers
- Level control / monitoring through a sight glass / non-metallic window
- Detecting / counting of small plastic / metal parts moving on a conveyor or through a non-metallic tube

## Typical Industries

- **Plastics:** Dryers, Vacuum systems, Hopper level control, conveyor systems
- **Agriculture:** Grain silos, automatic feeding systems, Irrigation systems
- **Wood Industry :** Window and door framing, saw dust / wood chip level control, wood pallet production
- **Material handling :** Wooden pallet detection on conveyors, Package inspection,
- **Food and Beverages:** Level detection in Bottles and packages, position control of packaging, Pallet detection

### Technical Specifications :

**Sensing range (Sd) 1.0 to 30 mm, factory set at 15 mm**

**Sensitivity Adjustable via multi-turn pot**

Effective operating distance (Sr)  $0.9 \times S_n < S_r < 1.1 \times S_n$

Usable operating distance ( $S_u$ )  $0.8 \times S_r < S_u < 1.2 \times S_r$

Repeat accuracy (R) &lt; 5%

Rated operational current ( $I_e$ ) < 100 mA (continuous)

No load supply current ( $I_o$ ) < 12 mA

Voltage drop ( $U_d$ ) < 2.5 VDC @ max. load

**Protection** Short-circuit, reverse polarity, transients

For statusYellow LED For AlarmRed LED For NormalBlue LED Environment

Operating temperature 0 to +60 deg. C

Housing material Body and check-nutSS

# RETRO REFLECTIVE OPTICAL LEVEL SWITCH

## Measuring Principal :

An infrared emitter send chopped light into the Prism. When in the air the light undergoes total internal reflection and is sensed by the Photo-detector. When the Prism is covered by the liquid, the light gets dispersed and so is not detected. This change is processed to signal an alarm.



## AppliCations :

Liquids like: Beer, Glucose , Alcohol, Petrol , Diesel ,Oil , Fruit Juices , Water , Chemicals , Honey Aerated Drinks, etc. Detection of Liquids in Pipelines Overfill and Dry Run Protection

## Typical Industries :

Breweries , water and waste Water Management , Petrochemicals, Food Processing , Pharmaceuticals , Cosmetics , Edible Oil, Chemicals , Machine Building.

## Specification:

Housing:SS316  
Probe Length: 35mm  
Process Fitting: 1/2", 3/4", 1" BSP /NPT  
Process Temperature: Upto 80 °C  
Ambient Temperature: -20°C to 80°C  
Output Type: PNP Standard  
Output:Max 100mA, Short – circuit protected  
Input Supply:12...30 VDC (30 mA max)

# ROTATING PADDLE POINT LEVEL SWITCH

## TS-ROTO 6000

|                         |                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------|
| Housing                 | : Cast Aluminium                                                                                               |
| Power supply            | : 230 V AC, $\pm 15\%$ or 110 V AC or 24 V DC                                                                  |
| Output                  | : SPDT 230 V AC, 10 A,                                                                                         |
| Ambient temperature     | : $-40\text{ }^{\circ}\text{C} \dots +65\text{ }^{\circ}\text{C}$                                              |
| Electrical Cable gland  | : PG -13, 2 nos. ( Polymer)                                                                                    |
| Insertion length        | : 150 mm standard,                                                                                             |
| Extension               | : Upto 3 mtrs                                                                                                  |
| Revolution per minute   | : 1 / minute Material of wetted parts : stainless steel                                                        |
| Number of vanes         | : stainless steel, 1 to 4, various types available as per application                                          |
| Process connection      | : 1, 1-1/2" BSP ( standard) others on request.                                                                 |
| Medium specific gravity | : min. 0,08 kg / dm <sup>3</sup>                                                                               |
| Medium temperature      | : $40\text{ }^{\circ}\text{C} \dots +80\text{ }^{\circ}\text{C}$ ,                                             |
| Special version         | : upto $260\text{ }^{\circ}\text{C}$ available                                                                 |
| Medium pressure max.    | : 2 bar                                                                                                        |
| Special Version         | : 1 Detection of solids under 10 mtr water column.<br>2 Angular probe for bulk material upto 25 mm grain size. |



### Theory :

A small electric motor drives the paddle, which rotates freely in the absence of material. Impeded by material the motor will be turned within the housing while loading a spring and triggering two switches. One of them is a dry electric contact for control and alarm functions while the other cuts the power to the motor. When the material level drops, the loaded stretched tension spring returns the motor to its original position and the unit is reactivated.

### APPLICATION

TSROTO suitable for level detection of powder, grains, granules and other free flowing solids. Mounted horizontally or vertically in bins, silos and hoppers.

### Main application areas :

|                   |                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------|
| Agriculture       | : beet slice, hard crop                                                                                                    |
| Chemical industry | : plastic powders, granules, pellets                                                                                       |
| Food industry     | : sun flower, sun flower cod,                                                                                              |
| Building industry | : sand, calcium powder, gypsum, coffee and cacao powder, flour, sugar, etc. Energetics: active soot, coal powder, fly ash. |

Number and material of vanes Specific gravity and particle size of the material provides orientation for the number of vanes. Most commonly used is the stainless steel, single vane blade. The lowest specific gravity for this paddle is 0,4 kg/dm<sup>3</sup>.

# ROTATING PADDLE LEVEL SWITCH –

## TS-ROTO 6240

|                          |                                                                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------------------|
| Housing                  | : Aluminium, weatherproof, powder coated.<br>Integral with the probe                                           |
| Cable Entry              | : 2nos.                                                                                                        |
| Mains Voltage            | : Universal 65 to 260 V AC , 50 Hz /24 VDC                                                                     |
| Fail safe                | : Selectable for High or Low                                                                                   |
| Output                   | : 2sets of potential free c/o contacts rated at 5 amps,<br>230 V AC for non-inductive loads.                   |
| Power Consumption        | : 8VA                                                                                                          |
| Mounting                 | : Screwed 1-1/2" BSP ( standard) Or Flanged(Optional)                                                          |
| Sense rod                | : Stainless steel, 12mm DI                                                                                     |
| Electrical Cable gland   | : PG -13, 2 nos. ( Polymer)                                                                                    |
| Insertion length         | :150 mm standard                                                                                               |
| Extension                | : Upto 1 mtrs                                                                                                  |
| Revolution per minute    | : 8 / minute                                                                                                   |
| Material of wetted parts | : stainless steel,                                                                                             |
| Number of vanes          | : stainless steel, 1 to 4,varioustypes available as per application.                                           |
| Process connection       | : 1, 1-1/2" BSP ( standard) others on request.                                                                 |
| Medium specific gravity  | : min. 0,08 kg / dm3                                                                                           |
| Medium temperature       | : 40 °C...+80 °C,                                                                                              |
| Special version          | : upto 260°C available                                                                                         |
| Medium pressure max.     | : 2 bar                                                                                                        |
| Special Version          | : 1 Detection of solids under 10 mtr water column.<br>2 Angular probe for bulk material upto 25 mm grain size. |



### Salient Features :

- No clutch, no gear , no spring.
- Flexible coupling (optional as per application).
- Robust, dust and watertight housing.
- Extended probe possible.
- Extended life by motor shut off design
- Torque adjustable low, medium & high as per medium.
- Sealed bearing.
- Interchangeable paddle assemblies.
- Fail safe switch selectable.
- 2 relay output each 1 SPDT.

### Function :

A stepper motor rotate the paddle slowly pulse counter coupled to the paddle shaft produces electrical pulses during its rotation. The pulses are monitored by a microcontroller to monitor its movement. Paddle rotation stops when the material covers it. The microcontroller gets signal from the encoder about material presence and signal is processed to derive potential free contacts. The microprocessor continuously checks the material presence. The paddle shakes and vibrates to discard the built up material has receded and paddle is free. Thereafter the output status is changed and motor starts rotating smoothly again waiting for the level to come.

### Application

TS-ROTO heavy duty , rotary paddle switches are the most suitable for level detection of dust suitable for level detection of dust granules and other free flowing solids up to the particle size of 10 mm. Mounted in bins, silos and hoppers it can be applied for control of level, filling and emptying as well as for protection against overfilling.

Primary areas of application : Building industry materials, cement, sand, lime, etc. Foodstuff industry , milk powder, flour, salt, food grains, etc. Plastics industry, powder, granular etc. Timber industry, chemical and mining.

## CONDUCTIVITY LEVEL SWITCH –



### TECHNICAL SPECIFICATIONS :

|                       |                                                                                                    |
|-----------------------|----------------------------------------------------------------------------------------------------|
| SWITCHING UNIT        | : Microcontroller Based.                                                                           |
| Housing               | : DINRAIL,suitableforback panel mounting                                                           |
| Ambient temperature   | : 0°Cto+60°C                                                                                       |
| Mains Voltage         | : 65-265VAC,50Hz/18-36V DC                                                                         |
| Output                | : 1 sets of potential free c/o contacts rated at 5 amps, 230VACforresistiveloads per switch point. |
| Power consumption     | : 2.5 VA                                                                                           |
| Safety operation      | : Field selected switchover for minimum or maximum(FSL/FSH)switching points.                       |
| Switch status display | : LEDL1toL4forfourpoints of level Alarm and L5 for power ON/ timer.                                |

### SENSING PROBE :

|                       |                                                                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mounting              | : Screwed – 1/2" BSP (standard), Above 1/2" BSP and flanged mounting (As per application and requirement) Sensing Probe: Stainless steel 304/ 316 rod / rope type |
| Insulation            | : PTFE (standard) Other insulation on request and as per application.                                                                                             |
| Operating temperature | : 200 ° C max.                                                                                                                                                    |
| Pressure              | : upto 40 Bar (Inside vessel)                                                                                                                                     |

### APPLICATIONS :

For conductive Liquids and Low-conductive Liquids such as DM-Water. Suitable for liquid having conductivity as low as 1 micro Siemens.

# DIAPHRAGM LEVEL SWITCH FOR SOLID-TSBOOT



## SPECIFICATIONS :

|                                  |                                                  |
|----------------------------------|--------------------------------------------------|
| Normal sensitivity               | : 200 gm (Rubber) / 400 gm (SS)                  |
| Diaphragm Membrane material      | : Nitrial Rubber / SS                            |
| Ambient temperature              | : 60 Deg                                         |
| contacts                         | : Micro switch, 15 Amp, 230 V AC, Potential free |
| Cable Gland                      | : PG 11                                          |
| Main body                        | : Cast Aluminum                                  |
| Switching Housing                | : Polymer for SS type                            |
| Protection for switching housing | : IP 67                                          |

## APPLICATION :

The TSBOOT level switch is used for indicating, controlling, and regulating the level of granulated or powdered dry materials with a grain diameter up to 30 mm and a density of 0,3 to 2,5 g/cm<sup>3</sup>, flowing through a silo, hopper or chute.

## OPERATION

The device must be mounted so that the diaphragm membrane is exposed to the product whose level should be controlled, in such a way that the product can press the diaphragm. The material weight presses the diaphragm which is adjusted with the micro switch through a flexible mechanism. A liver attached to the diaphragm transfers the pressure to the micro switch which changes the contact. When the material recedes the diaphragm comes back to its normal position and the micro switch again changes the contact.

## FEATURES

The TSBOOT consists of a micro switch inside the housing , covered by a flexible Diaphragm.

## INSTALLATION AND MOUNTING

The TSBOOT can be installed outside the container by making a hole on the wall, so that the membrane remains exposed to the product inside. Suspended by means of a tube, it can be also installed inside the container. Using adaptors. Wherever possible level switch should be mounted on vertical surfaces. By this means, the product can flow free downwards, and wrong indications are avoided. It can be mounted on surfaces with a slope of not more than 30 degrees from vertical, provided that the product l eaves the diaphragm membrane completely.

If maximum levels are to be controlled, the devices must be mounted at such a height that the product overflowing the membrane can exert pressure on this membrane, working the micro switch.

# FUEL / CAPACITANCE LEVEL TRANSMITTER

## Specifications : ( Microcontroller Based )

|                     |                                             |
|---------------------|---------------------------------------------|
| Housing             | : Cast Aluminium, weatherproof              |
| Mains               | : 12 V DC to 30 V DC                        |
| Current Consumption | : 20 mA(max)                                |
| Response Time       | : better than 10 sec                        |
| Operating Temp.     | : 20 to + 60 °C                             |
| Signal output       | : 0 to 5 V DC or 10 VDC , RS485/ 4-20 mA DC |
| Mounting            | : Screwed 3/4" BSP Or Flanged(Optional)     |
| Range Setting       | : Digital Setable                           |



## SENSING PROBE -

|                     |                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------|
| Mounting            | : Screwed 3/4" BSP(standard) Or Flanged(Optional)                                             |
| Sense               | : Pipe / Rod stainless steel (SS 304)                                                         |
| Grounding           | : Pipe, SS 304                                                                                |
| Length              | : 200 (standard) up to 2000 mm                                                                |
| Accuracy            | : For FUELTRANCE: 4 mm standard 2 mm special version upto 2000mm<br>For CAPTRNCE : +- 0.5% FS |
| Operating Temp in : | 70 deg for the integral unit (standard)                                                       |

## Application :

Fuel Level Sensor capacitance fuel trance is designed for diesel. Fuel level measurement in fuel tank of vehicles and stationary units. It is prohibited to use fuel trance for level measurement of electrically conducting liquids (e.g. water)

|                           |                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------|
| Conductive material       | : water based solutions acid based etc.<br>Only insulated probe are suggested.          |
| Non-conductivity material | : Mostly petrochemical and solvents nisulated<br>or non-insulated probes can be used.   |
| Liquid Application        | : Acid, Acitone, Amonia, Benzine, Feed water , Disel,<br>Oila, Fruit Juice, Milk, Wine. |

## Models :

- FUELTRANCE 5105 : OUTPUT: 0-5 V DC(3- Wire)
- FUELTRANCE 5110 : OUTPUT: 0-10 V DC(3-Wire)
- FUELTRANCE 5120 : OUTPUT: 4-20 mA DC(2 wire)
- FUELTRANCE 5458 : OUTPUT: Rs485 (4 Wire) & 0-5 V DC

## ULTRASONIC LEVEL TRANSMITTER FOR LIQUID:

### Specification :

|              |                                         |
|--------------|-----------------------------------------|
| Range        | : 5mtr,10mtr,15mtr,20mtr, 30mtr, 50 mtr |
| Power Supply | : -24V DC, 120 mA, 12 V D               |
| Analog O/P   | : 4-20 mA or Relay/Alarm output         |
| Digital O/P  | : -Rs485 (Optional)                     |
| Pressure     | : 2Bar (Explore Proof Type is 5 Bar)    |
| Accuracy     | : 0.2% --0.5% F.S.                      |
| Display      | : LCD Display                           |

### Application :

- water and wastewater treatment. Pumping Station, Collection wells, biochemical reaction tank, sedimentation tanks, etc. Electric power,mining Mortar pool, coal slurry pond, water treatment, etc.



## RADAR LEVEL TRANSMITTER:

|                    |                                |
|--------------------|--------------------------------|
| Application        | : Rivers, Lakes, Shoal         |
| Measuring Range    | : 70 meters                    |
| Process Connection | : Thread G1½ A? /Frame /Flange |
| Temperature        | : -40 °C ~ 100 °C              |
| Process Pressure   | : Normal pressure              |
| Precision          | : ± 3mm                        |
| Frequency Range    | : 26GHz                        |
| Protection Grade   | : IP67 / IP65                  |
| Power Supply       | : DC (6 - 24V) / Four-wire     |
| The Signal output  | : RS485 / Modbus Protocol      |
| The Scene Display  | : Optional                     |
| Shell              | : Aluminum / Plastic           |



# TDR RADAR LEVEL TRANSMITTER :

Radar Level Meter with Guided Wave (Principle TDR) ·

- Suited to continuous level measurement of various Liquid, mush and bulk-solid material.
- Universal use, direct mounting into containers, silos, vessels, reservoirs, etc.
- Stainless steel rod or rope electrode.
- Measuring range up to 40m.
- Linear measurement in non-conductive and differently shaped containers.
- Quick View measured values on the display.
- Simple installation and settings.
- Current output (4....20 mA), HART® protocol.

## Specification :

Supply Voltage : 8.....36V DC

Output : 4...20mA, HART ®

Basic error : 1) (for reference reflector)  $\pm 5\text{mm}/\pm 3\text{mm}/\pm 2\text{mm}$

Resolution : 1 mm

Maximal length of measuring electrode : 40m

Damping 1 ....99 s

Warm up time cca 60 s

Internal resistance / Electric strength (Electrode – Housing) 10 k $\Omega$

Coupling capacity / Electric strength (Housing – supply leads) 5nF / 500 V AC

Mounting:

## Application :

Rivers, Lakes, Shoal



## FLOW METERS

### ELECMAGNETIC FLOW METER

#### Specification :

|                                  |                                           |
|----------------------------------|-------------------------------------------|
| Nominal Dia(mm)                  | : 15 TO 2000                              |
| 2 Working pressure(kg/cm)        | : 10 to 40                                |
| Electrode Material               | : SS 316 standard                         |
| Sensor Lining                    | : Hard Rubber/ PTFE                       |
| Display Version                  | : Integral/Remote                         |
| Measuring tube material          | : SS304 Non Magnetic                      |
| Sesor housing material           | : CS / SS (Optional)                      |
| End Connection                   | : Flanged/Wafer/SMS                       |
| Flange                           | : Standard                                |
| Measuring range                  | : 0.2 mtr / sec to 12 mtr / sec           |
| Accuracy % of measured value     | : 0.5                                     |
| Repeatability                    | : 0.2 of span                             |
| Display                          | : 2 Line LCD                              |
| Output                           | : 4 to 20 mA DC/RS 485/PULSE Output /HART |
| Protection class for transmitter | : IP- 67                                  |
| Cable length for remote          | : 10 mtr (Optional)                       |
| Display Units                    | : All standard engineering Units          |
| Power Supply                     | : 24 V DC/80-300 V AC                     |
| Protection class for sensor      | : IP67                                    |
| Installation                     | : In Line Flanged Mounting                |
| Working temperature              | : Integral PTFE-120                       |
| 0                                | : Remote PTFE 0-180° C                    |
| 1                                | : Others-70 C                             |
| Optional                         | : IP 68 for flow tube in Remote Type      |



#### Application :

- Water Supply Network
- Chemical and Process Industries
- Pharmaceutical Industries
- Waste- Water Management
- Sugar, Food and Beverage Industries
- Effluent Treatment Plants

# MOISTURE SENSORS

## ON LINE MOISTURE SENSOR – FOR MIXERS AND SMALL VESSALS:

### Hydro-mix :

Hydro – Mix Flush mount, digital, microwave moisture sensor

### Specification :

#### Construction:

- Body : Stainless Steel
- Protection Ring : Hardened Steel
- Faceplate : Ceramic

Moisture Range : The sensor will measure upto saturation of material.

Material Penetration : Approximately 75-100mm dependent on material.

Fixing : The Hydro-Mix is secured to a fixing plate by a clamp ring. This allows the height of the sensor to be adjusted if necessary.

Operating Temperature : 0-60 degrees C. The sensor will not measure in ice.

Analogue Outputs : 2 Configurable 0-20mA or 4-20mA current loop source available for moisture and temperature. May also be converted to 0-10V DC.

Digital Inputs/Output : 2 configurable digital signals are provided for batch average start/stop or for moisture / temperature multiplexing purposes.

Connections : The sensor is fitted with a 10-way MIL-SPEC connector. available. A 4, 10 or 25 metre, 6 twisted pair cable pre-assembled for connection to a water proof junction box is available.

Power Supply : +15Vdc to +30Vdc. 4W

Digital (Serial) Communications : Opto-isolated RS485 2-wire port. Uses Hydro-com communications protocol for read/write access to sensor parameters and values. Ethernet and USB interfaces are also available.

### Features :

- Advanced Digital Technology
- Easy to Interface
- Wear Life
- Accuracy of Measurement
- Sensor Configuration
- Temperature Measurement
- Fully Temperature Compensated Measurement
- Measuring in the Mix

### Application :

- RMC
- PRECAST
- FREEFLOW CHEMICALS (POWDER)
- FOOD GRAINS
- SUGAR



# ON LINE MOISTURE SENSOR – HOPPER AND CONVEYOR:

## Hydro-Probe

### Specication

- Body : Stainless Steel
- Faceplate : Ceramic
- Refresh Rate : 25 times persecond.
- Fixing : The sensor must be placed in the ?ow of Material.
- Bins and Silos : Install in the neck of a bin or underneath the gate. Standardand Extension Mounting Sleeves are available to suit different bin widths.
- Conveyors : Secure in the ?ow ofmaterial.
- Operating Temperature : 0-60°C. The sensor will not measure ice.
- Moisture Range : The sensor will measure upto saturation of Material.
- Analogue Outputs : Two configurable 4-20mA or 0-20mA current loop source available for moisture and temperature. May also be converted to 0-10V DC.
- DigitalInputs/Output : 2 configurable digital signals available for averaging and alarm functions.
- Power Supply : +15V to +30V DC, 4W.
- Extension Cable : Six twisted pairs, 22AWG, 0.35mm 2 conductors. Screen braid with 65% minimum coverage plus Aluminum/polyester foil. Maximum cable runs of 100 m



### Features :

- Advanced Digital Technology
- Easy to Interface
- Wear Life
- Accuracy of Measurement
- Sensor Configuration
- Temperature Measurement
- Fully Temperature Compensated Measurement
- Measuring in the Mix

# DENSITY GAUGE MEASUREMENT METER:

## Specification :

### SOURCE :

Source Holder RTI 500mCi/ 18.5 Gbq (Cs-137) source holder ·

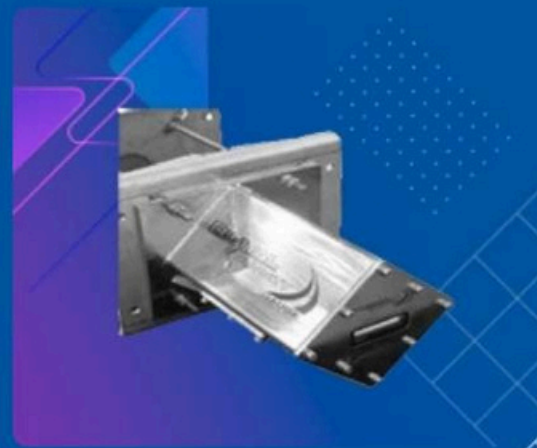
- All stainless steel construction (no lead fill) ·
- Drop tested and fire proof ·
- Rotary shutter ·
- Meets or exceeds internationally accepted safety standards Also compatible with most third party source holders (contact RTI for further information)
- Radioisotope : Double encapsulated Cs - 137 or Co - 60

### DETECTOR :

- Detector Element Sodium Iodide scintillation detector with integral photo multiplier. All electronics fully insulated into a single module.
- Housing IP67 316 stainless-steel, with stainless-steel mounting brackets with multiple mounting arrangements. Power 88- 264 V AC single phase , or 24 V DC. Data Integrity Radioactive Decay Clock.
- No data loss with power off.
- Communications & Control Output : 4-20 mA for SG (with separate 24V and 240 V connections)
- Setup/Control/Calibration/Diagnostics : iPad® via wireless communications\* (iPad® and Density Gauge App included)
- Mod Bus® also available
- Display Integral backlit LCD display on Detector Housing.
- Display SG and counts displayed for reference and basic diagnostics.
- Performance and Environmental : Temperature Drift  $\pm 0.00003$  SG units/ $^{\circ}\text{C}$
- Operating temperature: 0-80  $^{\circ}\text{C}$  Vibration 2.5 g at 100 Hz
- Humidity 5% - 95% RH non condensing

### Features:

- High sensitivity allows use of low activity sources.
- Built in protection from high count -rates (empty pipe)
- Rugged design , with built-in backlite LCD display
- Repeatability 0.0001 SG
- Automatic Gain Stabilization.



# Smart Level Sensors by Trivantas

Precision Monitoring for Liquids, Solids, Moisture & Gas

Wherever precision sensing is essential, Trivantas delivers. Our smart level sensors are engineered for accuracy and durability—available in waterproof, dust-proof, high-temperature, and high-pressure designs. We also specialize in custom-built sensors tailored to your application's exact requirements.

## 🔧 Why Choose Trivantas?

- ✓ Robust performance in demanding environments
- ✓ Waterproof, Dust-proof, High-temp & High-pressure options
- ✓ Tailor-made outputs to match your system requirements
- ✓ Application-specific modifications available

## 🌟 Product Categories 🌟

### 💧 Liquid Level Sensors & Switches

Accurate level sensing solutions for tanks, vessels, and pipelines.

- Float Type Level Sensors
- Vibrating Fork Level Sensors
- Conductivity Level Sensors
- Capacitance Level Sensors
- Fuel Level Transmitters
- Ultrasonic Level Sensors
- Radar Level Sensors
- Mini Radar Level Sensors
- Hydrostatic Level Sensors
- Magnetic Level Switches
- Side & Top Mounted Level Switches
- Miniature Level Sensors
- Level Transmitters
- Electromagnetic Flow Level Sensors

## **Solid Level Sensors & Switches**

Reliable detection of powders, granules, and bulk solids in hoppers and silos.

- Vibrating Fork Level Sensors
- Vibrating Rod Level Sensors
- NRF / RF Admittance Probes
- Rotating Paddle Level Sensors
- Capacitance Level Probes
- Microwave / Radar Solid Level Sensors
- Weight Level Switches
- Tilt Level Switches
- Mini Level Sensors
- Electromagnetic Flow Sensors

## **Moisture & Humidity Sensors**

Real-time measurements for process optimization and quality control.

- Online Moisture Sensors for Hoppers & Conveyors
- Bulk Material Moisture Measurement
- Capacitance-Based Moisture Sensors
- Infrared Moisture Analyzers
- Microwave Moisture Sensors

## **Gas Level & Detection Sensors**

Smart monitoring of gas flow, pressure, and concentration levels.

- Gas Pressure Level Sensors
- Ultrasonic Gas Flow & Level Sensors
- Capacitance-Based Gas Detection Probes
- Thermal Mass Flow Level Sensors

## **Custom Solutions Available**

Need a sensor adapted to your system? Trivantas offers custom design & output configurations to perfectly match your industrial needs.